

1. Introduction

This project demonstrates the implementation of an end-to-end DevOps workflow, including containerization, service orchestration, cloud deployment, and CI/CD automation.

The application consists of:

- A **FastAPI backend**
- A **static frontend served via Nginx**

The objective was to build, deploy, and automate the application using modern DevOps practices.

2. Project Setup

Backend Setup

The backend was developed using FastAPI and tested locally using:

```
python -m uvicorn app.main:app --reload
```

This verified that the application was functioning correctly before containerization.

Frontend Setup

The frontend is a static HTML interface served using Nginx.

3. Dockerization

Backend

The backend was containerized using a Python base image and executed using Uvicorn.

Frontend

The frontend was containerized using Nginx:

```
FROM nginx:alpine  
COPY . /usr/share/nginx/html  
EXPOSE 80
```

This ensured lightweight and efficient serving of static content.

4. Docker Compose Orchestration

A docker-compose.yml file was created to orchestrate both services.

Configuration Highlights:

- Backend exposed on port **8000**
- Frontend exposed on port **80**
- Volume configured for persistence

Command Used:

```
docker-compose up --build
```

This allowed both services to run simultaneously with proper networking.

5. Cloud Deployment (AWS EC2)

The application was deployed on an AWS EC2 instance using Ubuntu.

Steps Performed:

- Launched EC2 instance
- Configured security group:
 - Port 22 (SSH)
 - Port 80 (Frontend)
 - Port 8000 (Backend)
- Installed Docker and Docker Compose
- Cloned the GitHub repository
- Deployed application using:

```
sudo docker-compose up -d --build
```

Access URLs:

- Frontend: `http://<EC2-PUBLIC-IP>`
- Backend: `http://<EC2-PUBLIC-IP>:8000`

6. CI/CD Pipeline

A CI pipeline was implemented using GitHub Actions.

Workflow:

- Triggered on push to main branch
- Code checkout
- Docker image build
- Container execution
- Application validation using curl

This ensures continuous integration and automatic verification of application functionality.

7. Challenges Faced

During the implementation, several real-world issues were encountered:

- **pip not recognized**
→ Resolved using `python -m pip`
- **Module import error**
→ Fixed using module execution (`-m` flag)
- **Docker engine not running**
→ Resolved by starting Docker Desktop
- **GitHub permission denied (403)**
→ Fixed by forking repository and updating remote
- **CI/CD not triggering**
→ Corrected workflow directory to `.github/workflows`
- **Docker Compose not available on EC2**
→ Installed manually using binary method

8. Conclusion

This challenge provided hands-on experience in:

- Docker containerization
- Multi-service orchestration using Docker Compose
- Cloud deployment using AWS EC2
- CI/CD pipeline implementation using GitHub Actions
- Debugging real-world DevOps issues

The project successfully demonstrates a practical understanding of deploying and automating applications using DevOps methodologies.

The screenshot displays a development environment in VS Code. The Explorer sidebar on the left shows a project structure for 'fusionpack-devops-challenge' with folders like '.env', '.git', 'hooks', 'info', 'logs', 'objects', 'refs', 'config', 'description', 'HEAD', 'index', 'packed-refs', 'backend', 'app', 'requirements.txt', 'frontend', 'DevOps_Intern.html', 'README.md', and 'SOP CREATE HOME WEBPAGE USING NGINX...'. The Dockerfile in the main editor contains the following content:

```
1 FROM python:3.9
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 8000
11
12 CMD ["python", "-m", "uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

The terminal at the bottom shows the execution of Docker commands:

```
PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpack-devops-challenge\backend> docker build -t fusion-backend .
=> exporting attestation manifest sha256:2b6f4dc4a9bd44a6718b8170776fd8d81bd89a759580eaa41b4341fcf026c6 0.0s
=> exporting manifest list sha256:db9feca015ae111a2f4c5c8c0ba2a7a5846e2fae11eab9e42753cf4810d735c 0.0s
=> naming to docker.io/library/fusion-backend:latest 0.0s
=> unpacking to docker.io/library/fusion-backend:latest 0.3s
PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpack-devops-challenge\backend> docker run -p 8000:8000 fusion-backend
INFO: started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.17.0.1:47262 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:47262 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 172.17.0.1:39660 - "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 Not Found
```

Below the terminal, a web browser window shows the local host at port 8000. The browser's developer tools console displays the following message:

```
{ "message": "Hello from FastAPI - @klrnmrakh155@gmail.com :)" }
```

```
1 name: DevOps CI Pipeline
2
3 on:
4   push:
5     branches: [ "main" ]
6
7 jobs:
8   build:
9     runs-on: ubuntu-latest
10
11     steps:
12     - name: Checkout code
13       uses: actions/checkout@v3
14
15     - name: Build Docker Image
16       run: docker build -t fusion-backend ./backend
17
18     - name: Run Container (Test)
19       run: |
20         docker run -d -p 8000:8000 fusion-backend
21         sleep 5
22         curl http://localhost:8000
```

PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpack-devops-challenge> mkdir .github

Mode LastWriteTime Length Name

d----- 07-05-2026 03:31 workflows

PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpack-devops-challenge> git add .

>> git commit -m "Final CI pipeline setup"

>> git push

[main dabb6be] Final CI pipeline setup

1 file changed, 22 insertions(+)

create mode 100644 .github/workflows/ci.yml

Enumerating objects: 6, done.

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#).

Actions

[New workflow](#)

All workflows

DevOps CI Pipeline

Management

- [Caches](#)
- [Attestations](#)
- [Runners](#)
- [Usage metrics](#)
- [Performance metrics](#)

All workflows

Showing runs from all workflows

[Filter workflow runs](#)

[Help us improve GitHub Actions](#)

Tell us how to make GitHub Actions work better for you with three quick questions. [Give feedback](#)

1 workflow run

Workflow ▾ Event ▾ Status ▾ Branch ▾ Actor ▾

Final CI pipeline setup

DevOps CI Pipeline #1: Commit dabb6be pushed by Somitraa

main

1 minute ago

39s

← → ↻ 🔍 localhost/#fulltime-offer

Gmail

WhatsApp

Google

speedTEST

YouTube

ChatGPT

Your projects - Over...

GitHub

PDF

JioHotstar - Watch...

Claude

All Bookmarks

Fusionpact DevOps Internship Program 2026

Become Industry-Ready DevOps Professionals | Hands-on Cloud, Containers, CI/CD & Monitoring

Month 1

Month 2

Month 3

Month 4

Month 5

Month 6

Full-Time Offer

Month 1 – Foundations

Linux CLI, Bash Scripting, Git Version Control

Learn & Practice:

- Master Linux navigation & commands
- Automate tasks using Bash scripts
- Setup Git repositories & track projects
- [Linux](#) [Linux CLI Guide](#)
- [Git](#) [Git Documentation](#)
- Real-world problem: Automate daily file backup

Month 2 – Cloud & IaC

AWS/GCP, Terraform Modules

Learn & Practice:

- Provision virtual machines & storage using Terraform
- Deploy static websites with infrastructure as code
- [AWS](#) [AWS Training](#)
- [Terraform](#) [Terraform Tutorials](#)
- Real-world problem: Automate infrastructure provisioning for multiple environments

Month 3 – Containerization

Docker Images & Compose

Learn & Practice:

- Containerize applications & databases
- Setup multi-container environments with Docker Compose
- [Docker](#) [Docker Guide](#)
- Real-world problem: Containerize a web + DB stack and deploy

Month 4 – CI/CD

Jenkins Pipelines, Automated Builds

Learn & Practice:

- Build CI/CD pipelines

Month 5 – Monitoring & Logging

Prometheus, Grafana, ELK Stack

Learn & Practice:

- Setup monitoring & logging

Month 6 – Capstone

Microservices, Docker, Kubernetes, CI/CD

Learn & Practice:

- Deploy multi-service microservices

Dockerfile ...\backend X ci.yml Dockerfile ...\frontend .gitignore docker-compose.yml

fusionpact-devops-challenge > backend > Dockerfile > ...

```
1 FROM python:3.9
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 8000
11
12 CMD ["python", "-m", "uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS docker-

```
PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpact-devops-challenge> docker-compose up --build
#20 [frontend] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 6/6
✓ Image fusionpact-devops-challenge-frontend Built
✓ Image fusionpact-devops-challenge-backend Built
✓ Network fusionpact-devops-challenge_default Created
✓ Volume fusionpact-devops-challenge_data Created
✓ Container fusionpact-devops-challenge-backend-1 Created
✓ Container fusionpact-devops-challenge-frontend-1 Created
Attaching to backend-1, frontend-1
frontend-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform
frontend-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default
```

Dockerfile ...\backend X ci.yml Dockerfile ...\frontend .gitignore docker-compose.yml

fusionpact-devops-challenge > backend > Dockerfile > ...

```
1 FROM python:3.9
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6 RUN pip install --no-cache-dir -r requirements.txt
7
8 COPY . .
9
10 EXPOSE 8000
11
12 CMD ["python", "-m", "uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS docker-

```
PS C:\Users\ASUS\Desktop\FUSION-PACT\fusionpact-devops-challenge> docker-compose up --build
#20 [frontend] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 6/6
✓ Image fusionpact-devops-challenge-frontend Built
✓ Image fusionpact-devops-challenge-backend Built
✓ Network fusionpact-devops-challenge_default Created
✓ Volume fusionpact-devops-challenge_data Created
✓ Container fusionpact-devops-challenge-backend-1 Created
✓ Container fusionpact-devops-challenge-frontend-1 Created
Attaching to backend-1, frontend-1
frontend-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform
frontend-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default
```


